

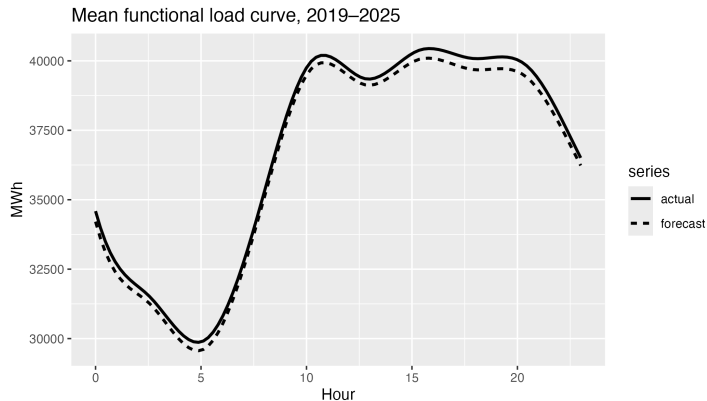
Functional Forecasting of Turkish Electricity Load Curves

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Thesis Proposal

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The Forecasting Object: A Daily Load Curve



$$Y_d(h), \quad h = 0, \dots, 23$$

- A single **curve**, not 24 scalars
- Dual-peak shape is stable across all days
- Solid = actual; dashed = official forecast
- **Goal:** forecast tomorrow's full 24-hour profile

Why Accuracy Matters

Key users (post-2013 liberalisation):

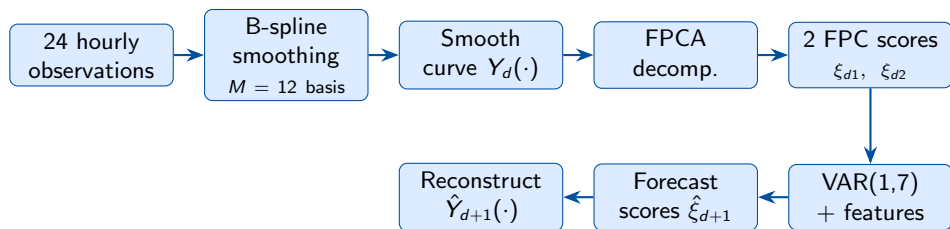
- **TEİAŞ** (system operator)
unit commitment, grid scheduling
- **Generators**
day-ahead market bids on EPIAS
- **Large consumers**
real-time imbalance penalties

Systematic underprediction — official TEİAŞ forecast

Year	Official bias (MWh/hour)	Our bias (MWh/hour)
2023	+344	+176
2024	+585	+34
2025	+472	+98
Average	+467	+103

Positive bias = forecast < actual (underprediction).

From 24 Numbers to a Curve: The FDA Pipeline



B-splines

preserves shape; enables differentiation

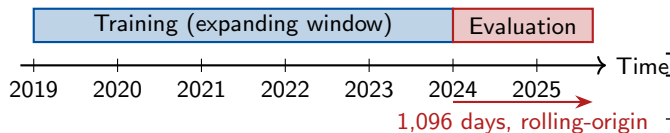
FPCA

2 scores carry 97.3% of curve variation

VAR(1,7)

weekly + daily score dynamics

Data



FPCA and VAR re-estimated before each forecast day.
Only data dated $< d$ used to forecast day d .

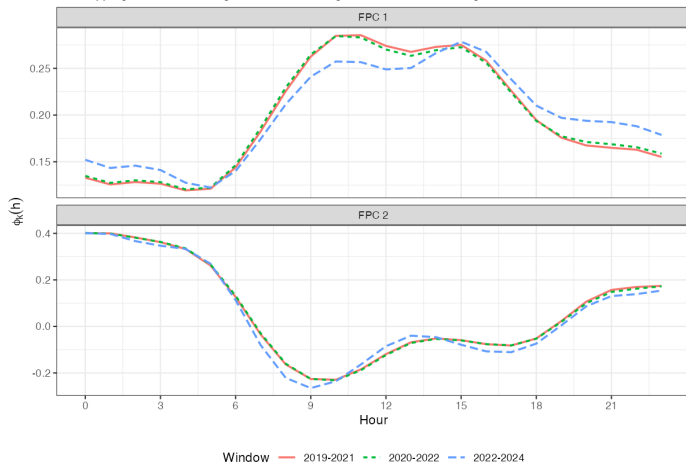
Series	Source
Hourly load (MWh)	EPIAS
Official forecast	TEİAŞ / EPIAS
Holiday calendar	Official gazette

Item	Value
Frequency	Hourly
Coverage	2019–2025
Training curves	$\approx 1,460$ days
Evaluation curves	1,096 days
Hourly errors eval	26,304

FPCA: Two Numbers Capture the Curve

FPCA eigenfunctions across sub-periods

Overlapping lines = stable eigenfunctions; divergence = structural change



FPC	Variance	Cumulative
1	93.4%	93.4%
2	3.9%	97.3%
3	1.1%	98.4%

FPC 1 — Level

Nearly flat $\Rightarrow$ shifts the whole curve up or down

FPC 2 — Shape

Positive overnight & late evening, negative daytime $\Rightarrow$ overnight-vs-daytime contrast

Stability: three sub-period fits nearly coincide $\Rightarrow$ static FPCA justified

Score Forecasting Model

Core score dynamics:

$$\xi_d = a + A_1 \xi_{d-1} + A_7 \xi_{d-7} + B z_d + u_d$$

- **Lag 1:** yesterday's level and shape
- **Lag 7:** same-day-last-week seasonality
- z_d : feature vector (all predetermined)

Reconstruction:

$$\hat{Y}_d(h) = \hat{\mu}(h) + \hat{\phi}_1(h) \hat{\xi}_{d1} + \hat{\phi}_2(h) \hat{\xi}_{d2}$$

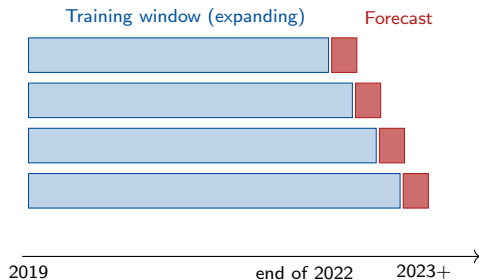
Estimated by OLS, equation-by-equation.

Feature blocks in z_d :

Block	Content
Calendar	Weekend, annual sine/cosine
Holiday	Public holidays, Eid, bridge days
Load shape	Yesterday's mean, peak, range
Rolling avg.	7-day, 28-day averages
Derivative	Morning ramp, evening ramp, slope volatility

Holiday and derivative blocks are the critical additions — without them the model does not beat the official.

Evaluation: Rolling-Origin Design



Each forecast uses only data dated before that day.
FPCA and VAR both re-estimated at every step.

Metrics:

$$\text{MAE} = \frac{1}{N} \sum_{d,h} |e_{dh}|$$

$$\text{MAPE} = \frac{1}{N} \sum_{d,h} \frac{|e_{dh}|}{Y_d(h)}$$

Significance test:

Diebold-Mariano (1995)

H_1 : challenger MAE < official MAE

Pooled (26,304 obs.) + hour-by-hour

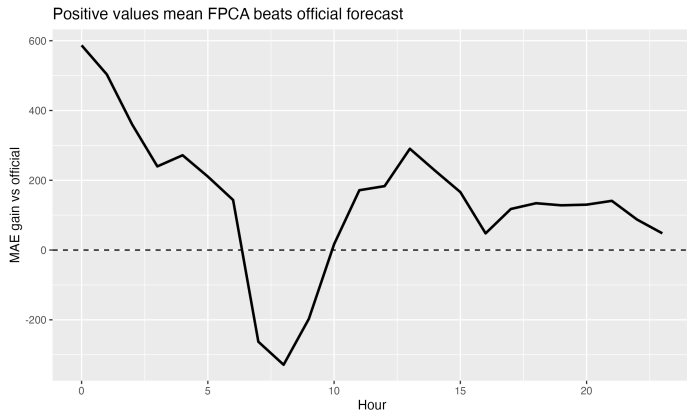
A lower MAE number alone is not sufficient evidence.

Main Results

Model	MAE	MAPE	vs. official	DM
Seasonal naïve Y_{d-7}	1912	5.13%	-57.6%	—
Official (TEİAŞ)	1214	3.15%	ref.	—
FPCA VAR(1,7)	1443	3.76%	-18.9%	n.s.
+ calendar / ramp	1238	3.26%	-2.1%	n.s.
+ holiday / load-shape	1001	2.63%	+17.5%	***
+ derivative features	980	2.56%	+ 19.2%	***
+ residual correction	653	1.70%	+ 46.2%	***

*** $p < 0.001$ (DM test, pooled over 26,304 obs.). MAE in MWh/hour. Green = beats official.

Where the Model Wins and Where It Loses



Positive = model beats official (MAE, MWh)

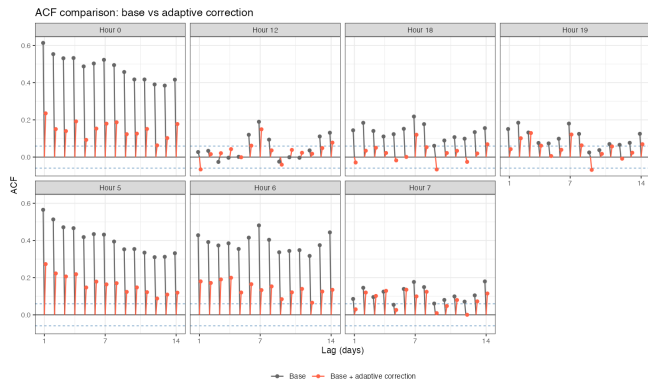
Wins: hours 0–5 and 10–23
(overnight trough and daytime)

Loses: hours 6–9
(morning ramp-up)

The source of the official forecast's ramp-hour advantage is not known. Our model uses historical load patterns only.

DM: significant at 18 of 24 hours (5% level).

Residual Correction: Exploiting Persistent Errors



Motivation: errors are serially correlated
Ljung-Box rejects at all 24 hours

$$\hat{Y}_d^{\text{res}}(h) = \hat{Y}_d^{\text{base}}(h) + \lambda_d \hat{r}_d(h)$$

$\hat{r}_d(h)$: correction curve from recent residuals

λ_d : rolling CV on pre- d data only

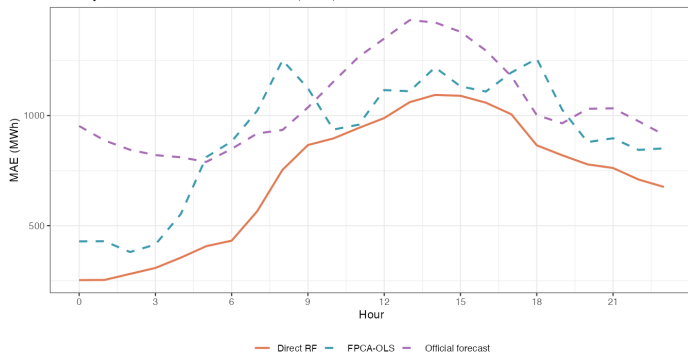
Result:

- +46% MAE vs. official
- Wins at **all 24 hours** (incl. ramp hours)

Gray = base model ACF; Orange = after correction

Robustness Checks

MAE by hour: Direct RF vs FPCA-OLS (2023)



Direct RF vs. FPCA-OLS (2023):

- Direct RF (no FPCA) wins at most daytime hours (8–23)
- FPCA-OLS wins overnight (0–5)
- Neither model uses temperature data

$K = 4$ components:

K	MAE	MAE (+resid.)
2	980	653
4	801	640

$K = 2$ is the parsimonious baseline.

Thesis Road Map

Completed

- ✓ Functional representation (B-splines, FPCA)
- ✓ Rolling-origin evaluation framework
- ✓ VAR(1,7) + feature blocks
- ✓ Residual correction step
- ✓ Diebold-Mariano testing (pooled + hourly)
- ✓ Random forest benchmark
- ✓ Eigenfunction stability analysis, $K = 4$ check

Planned

- ☐ Systematic literature review
- ☐ Weather/temperature covariates (ramp hours)
- ☐ Probabilistic forecasting (prediction intervals)
- ☐ Thesis write-up

References

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